IRIS RATIOS

★ *Denotes most likely to be tested*
If PH Surplus <= 0, then ratios are 999.

	RATIO	FORMULA	RANGE	INTERPRETATION	OTHER NOTES
OVERALL	★ 1: GWP to PH Surplus	(Direct WP + Reins Assm (Aff + Non-Aff)) / PH Surplus	≤900%	Measures adequacy of cushion to absorb losses (ignoring ceded premiums). high value = more risk in relation to surplus	Recalc if IRIS 4 is unusual Don't want gap between Ratios 1&2 to be too small or too large
OVERALL	★ 2: NWP to PH Surplus	NWP / PH Surplus	≤300%	Measures adequacy of cushion to absorb losses (net of ceded premiums) high value = more risk in relation to surplus	Recalc if IRIS 4 is unusual Don't want gap between Ratios 1&2 to be too small or too large
OVERALL	★ 3: Change in NWP	(CY NWP - PY NWP) / PY NWP	-33% to +33%	High or low indicates a lack of stability in operations High ratio could mean less strict underwriting req or writing new line	Familiarity with insurer (expanding in new area or writing new lob?) is helpful when interpreting
OVERALL	★ 4: Surplus Aid to PH Surplus	Surplus Aid / PH Surplus *(See other side)*	<15%	High = PH surplus may be inadequate. High surplus aid may conceal important results on other ratios	Must recalc IRIS 1, 2, 7, 10 and 13 with surplus aid removed from surplus if IRIS 4 is unusual.
PROFITABILITY	5: 2yr Op Ratio	2 Yr Loss Ratio + 2 Yr Expense Ratio - 2 Yr Inv Income Ratio *(See other side)*	<100%	Low = better operating profit helps determine why operating profit could be bad - expenses, losses or investment	Recalc if IRIS 11 is unusual
PROFITABILITY	6: Investment Yield	2 x NII Earned / (Total Cash + Invested Assets CY & PY + Inv Income Due & Accrued CY & PY - Borrowed Money CY & PY - NII Earned)	2% to 5.5%	Too low = speculative instruments giving capital gains with no interim income Too high = high risk instruments	
PROFITABILITY	7: Gross Change in PH Surplus	(PH Surplus CY - PH Surplus PY) / PH Surplus PY	-10% to 50%	Low = dangerous surplus decrease (may be caused by decrease in net income) High = possible insolvency (surplus often goes up before insolvency)	Recalc if IRIS 4 is unusual Large increase may indicate instability/upcoming insolvency
PROFITABILITY	8: Change in Adj PH Surplus	(PH Surplus CY - Chng in Surplus Notes - Capital Pd in or transferred - Surplus Pd in or transferred - PH Surplus PY) / PH Surplus PY	-10 to 25%	Low = deterioration in financial condition due to operations High = improvement in financial condition due to operations	Helps highlight insurer's actual operations
LIQUIDITY	9: Adj Liabilities to Liquid Assets	(Total Liabilities - Liabilities Equal to deferred agents' balances) / [(Bonds + Stocks + Cash, Cash Eqv & Short Term Inv + Receivables for Securities + Inv Inc Due and Accrued) - Invest in Parent, Subsid & Affiliates]	<100%	High ratio means an insurer might have trouble meeting short term obligations	Increasing trend is a bad sign
LIQUIDITY	10: Gross Agents Bal to PH Surplus	Gross Agents' Balances in Collection / PH Surplus	<40%	High ratio means that an agent might be slow in paying	Recalc if IRIS 4 is unusual Balances > 90 days overdue may need to be removed from admitted assets
RESERVES	★ 11: One Yr Reserve Dev to PH Surplus	1 Yr Loss Reserve Development / (PH Surplus PY)	<20%	Positive = reserve deficiency; Negative = reserve redundancy	Can isolate LOB/AY using Sch P Pt 2
RESERVES	★ 12: Two Yr Reserve Dev to PH Surplus	2 Yr Loss Reserve Development / (PH Surplus 2nd PY)	<20%	Positive = reserve deficiency; Negative = reserve redundancy	Can isolate LOB/AY using Sch P Pt 2
RESERVES	★ 13: Est Curr Reserve Deficiency to PH Surplus	Estimated Reserve Deficiency* / PH Surplus *(See other side)*	<25%	Positive = reserve deficiency; Negative = reserve redundancy	Recalc if IRIS 4 is unusual Affected by changes in mix or prem volume Good test for correction of reserve deficiencies

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IRIS 4

Surplus Aid = ((Reinsurance Ceded Commissions + Reinsurance Ceded Contingent Commissions) / (Reinsurance Premiums Ceded - Affiliates and Non-Affiliates)) x (Unearned Premiums - Total Authorized, Unauthorized, Certified and Reciprocal Jurisdiction for Other US Unaffiliated Insurers, Mandatory & Voluntary Pools and Other Non-US Insurers)

IRIS 5

$$\frac{\text{Loss \& LAE CY \& PY + Dividends to PH CY \& PY}}{\text{EP CY + PY}} + \frac{\text{Other UW Expenses CY \& PY - Other Income CY \& PY}}{\text{WP CY + PY}} - \frac{\text{Investment Income Earned CY \& PY}}{\text{EP CY + PY}}$$

IRIS 13

A = 2nd PY Developed Loss & LAE Reserves to Premiums Ratio = (2nd PY Loss & LAE Reserves + 2 Yr Loss Reserve Development) / 2nd PY Premiums Earned
B = 1st PY Developed Loss & LAE Reserves to Premiums Ratio = (1st PY Loss & LAE Reserves + 1 Yr Loss Reserve Development) / 1st PY Premiums Earned
Estimated Loss & LAE Reserve Deficiency = (((A+B)/2) x CY Premiums Earned) - CY Loss & LAE Reserves

my notes:

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